LETTER TO THE EDITOR

Paleomagnetism and *K-Ar* Ages of Successive Lava Flows (3)*

——Hanamagari and Kirizumi andesites, Gumma Prefecture, Japan——

(Received February 27, 1968)

The northeastern part of Nagano Prefecture and western part of Gumma Prefecture belong to the extensive Miocene-to-Quaternary volcanic province of the "Fossa Magna" in central Japan (Fig. 1). A partly destroyed Quaternary strato-volcano, Hanamagari (*Aramaki*, 1963) lies on the boundary between Nagano and Gumma Prefectures and is underlain by a thick pile of volcaniclastic materials and lava flows. This pile is collectively called the Kirizumi group (*Iijima et al.*, 1958; *Aramaki*, 1963). The Kirizumi group is more or less flat-lying and made up of essential tuff, lapilli tuff, tuff breccia, lava flow, and smaller amounts of sandstone, shale, conglomerate, etc. A few geological evidences indicate that the Kirizumi group is Pliocene. The rocks of both the Hanamagari volcano and the Kirizumi group are calc-alkali andesites with common phenocrysts of plagioclase, hypersthene, augite, and opaque minerals (*Aramaki*, 1963).

Within the drainage area of the Kirizumi River, both the Hanamagari and Kirizumi

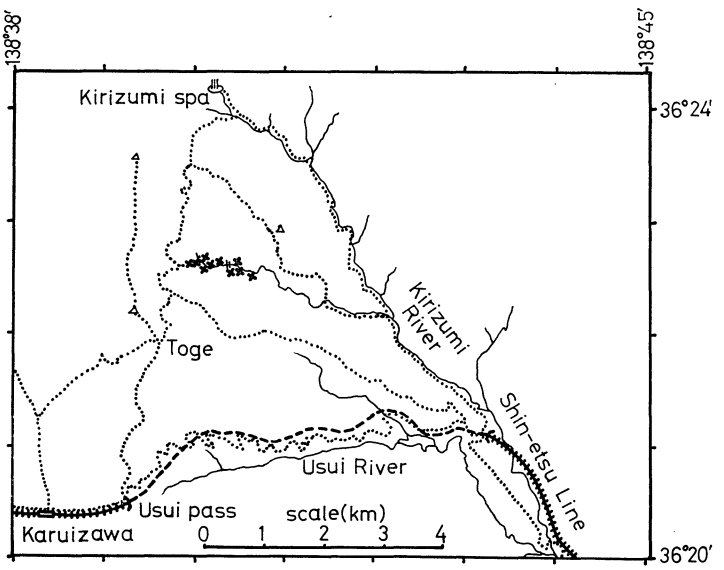


Fig. 1 Geological map of sampling site. Sampling sites are indicated by crosses.

* Contribution No. 241 from Division of Geomagnetism and Planetary Physics, Geophysical Institute, University of Tokyo, Series II.

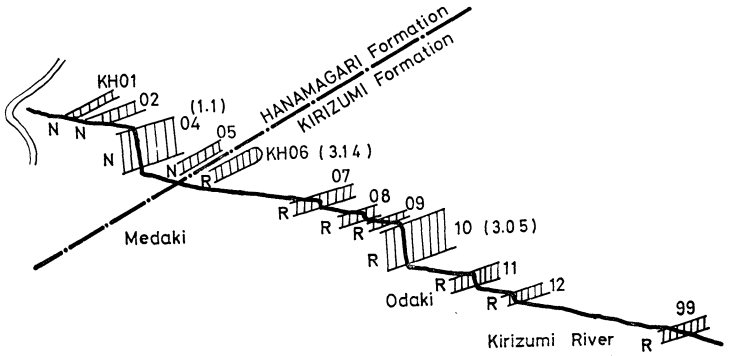


Fig. 2 Schematic representation of lava flow successions. N and R indicate normal and reversed polarities of NRM's in each flow. Mean *K-Ar* ages (m.y.) are also indicated.

pyroclastics and lavas are exposed and their stratigraphic succession can be clearly traced. Along one of the branch stream flowing from west to east to join the main course of the Kirizumi River, five successive lava flows of the Hanamagari volcano are underlain uncomformably by more than 15 lava flows of the Kirizumi group. Sampling was made on these lava flows as shown schematically in Fig. 2. The lava flows are intercalated with loose to semi-consolidated pyroclastic materials. The altitude of the boundary surface of the unconformity between the Hanamagari volcano and Kirizumi group changes more than 50 m within the horizontal distance of 300 m. The top of the Kirizumi group has been weathered to the depth of 10 m or more. The alternation of lava flows and loose pyroclastic materials of the Hanamagari volcano bears features characteristic to volcanic products of subaerial origin. The volcaniclastic materials intercalated with lava flows of the Kirizumi group are variagated, more or less equigranular tuff, lapilli tuff and tuff breccia with matrix of finely divided materials of the same nature as those composing coarser fractions. Many outcrops are interpreted as part of autobrecciated lava or thick mantle of brecciated part of the lava flow. All of these features indicate that the Kirizumi group is of subaqueous (submarine) origin and the Hanamagari volcano subaerial, and a large time gap existed between the two.

The interface between lava flows and volcaniclastic layers of the Kirizumi group appears to be always conformable indicating successive deposition within a short period.

3–5 cores were drilled from each lava flow. After measuring the NRMs, all samples were subjected to *a-c* demagnetization with a peak field up to 400 *Oe*. All flows belonging to the Hanamagari group have a normal polarity. Since the grouping of the NRMs is generally good and there is no marked change in the NRMs after the *a-c* demagnetization (Table 1), we conclude that the NRMs of the Hanamagari lava flows represent the paleomagnetic field, or the geomagnetic field had a normal polarity when the Hanamagari lavas were erupted. In contrast to the Hanamagari lavas, the lava flows of the underlying Kirizumi group have either a very shallow positive inclination or negative inclination. Although the NRMs before and after the *a-c* demagnetization have poor grouping, all the NRMs show

Table 1. Paleomagnetic data of Hanamagari and Kirizumi lava flows

Sample	*a-c* demagnetization*	NRM					Virtual pole		Intensity (e.m.u/gr)
		N	*I*	*D*	*k*	α_{95}	lat.	long.	
KH01	$\tilde{H}=o$	7	47.1	0.1	56.4	8.1	81.9	318.3	2.14×10^{-3}
	$\tilde{H}=200\ Oe$	7	47.4	1.7	45.4	9.1	82.1	308.0	1.19×10^{-3}
KH02	$\tilde{H}=o$	6	55.1	8.6	33.9	11.7	83.0	231.8	1.06×10^{-3}
	$\tilde{H}=200\ Oe$	6	59.9	8.5	20.6	15.1	82.0	192.8	6.49×10^{-4}
KH04	$\tilde{H}=o$	6	84.8	−136.7	3.61	41.1	28.5	130.6	5.89×10^{-4}
	$\tilde{H}=200\ Oe$	6	73.6	24.1	3.71	40.3	62.1	164.9	1.41×10^{-4}
KH06	$\tilde{H}=o$	5	30.4	−21.5	1.92	78.1	62.3	8.0	1.49×10^{-3}
	$\tilde{H}=200\ Oe$	5	−35.3	−61.3	1.70	89.7	9.6	15.7	4.07×10^{-4}
KH07	$\tilde{H}=o$	6	29.1	−50.3	1.59	84.5	40.9	37.3	1.90×10^{-4}
	$\tilde{H}=200\ Oe$	6	−56.5	−79.4	1.54	88.5	−13.8	12.6	1.10×10^{-4}
KH08	$\tilde{H}=o$	5	10.6	−45.5	3.28	50.2	38.1	23.2	1.18×10^{-4}
	$\tilde{H}=200\ Oe$	5	−45.0	−46.0	2.65	58.9	13.6	0.1	6.24×10^{-5}
KH09	$\tilde{H}=o$	5	66.7	−60.0	2.05	73.3	45.5	85.1	2.22×10^{-4}
	$\tilde{H}=200\ Oe$	5	−25.5	−118.8	1.98	75.9	−31.0	54.8	1.55×10^{-4}
KH10	$\tilde{H}=o$	5	66.5	−25.2	32.4	13.6	67.7	91.2	1.83×10^{-4}
	$\tilde{H}=200\ Oe$	5	−53.2	−112.0	9.41	26.3	−35.5	29.8	6.37×10^{-5}
KH11	$\tilde{H}=o$	5	−9.5	−96.5	2.29	66.4	−8.1	48.7	1.73×10^{-4}
	$\tilde{H}=200\ Oe$	5	−55.0	−138.0	10.5	24.8	−56.3	37.3	2.29×10^{-4}
KH12	$\tilde{H}=o$	5	−63.9	−151.5	82.6	8.5	−66.7	16.5	8.31×10^{-4}
	$\tilde{H}=200\ Oe$	5	−64.5	−142.7	53.6	10.5	−60.6	17.0	8.00×10^{-4}
KH99	$\tilde{H}=o$	2	−17.3	−174.3	11.6	81.6	−62.0	126.7	1.24×10^{-4}
	$\tilde{H}=200\ Oe$	2	−52.7	−179.6	880.6	8.4	−86.9	132.5	1.14×10^{-4}

* $\tilde{H}=o$ denotes an original NRM. $H=200$ denotes the NRM after *a-c* demagnetization with $\tilde{H}=200\ Oe$.

N number of samples.

I Inclination of mean direction of NRM.

D Declination of mean direction of NRM.

k Fisher's precision parameter.

α_{95} Semiangle of cone of 95% confidence for mean direction.

a negative inclination after the *a-c* demagnetization. Hence, it is likely that the geomagnetic field was reversed at the time the Kirizumi group was deposited. As will be discussed below, this conclusion is further substantiated by the comparison of the *K-Ar* ages with N-R geomagnetic polarity time table (*Dalrymple et al*, 1967, *McDougall* and *Chamalaun*, 1966). The results of the paleomagnetic measurements are listed in Table 1.

Thin sections were examined for all the lava flows. Except for flows KH04, KH06 and KH 10, they carry considerable amounts of alteration of minerals such as chlorite, montmorillonite etc. Hence, *K-Ar* dating was attempted only for the three flows, KH04, KH06 and KH10. Petrographic descriptions of these flows are given in the appendix. The results

Table 2. *K*-Ar ages of Hanamagari and Kirizumi lava flows

Sample	wt (gr)	(Ar^{40}) rad moles/gr	$\frac{(Ar^{40}) \text{ air}}{(Ar^{40}) \text{ total}}$	*K* (%)	*t* (m.y.)
KH04-5	7.685	1.164×10^{-12}	97.1	0.650, 0.643	1.01
	6.942	1.405×10^{-12}	96.5		1.22
KH06-2	8.787	4.416×10^{-12}	74.5	0.806, 0.797	3.10
	8.985	4.573×10^{-12}	85.1		3.17
KH10-3	13.42	5.078×10^{-12}	67.6	0.935, 0.936	3.04
	11.39	5.122×10^{-12}	73.5		3.07

$\lambda_e=0.585\times10^{-10}y^{-1}$, $\lambda_\beta=4.72\times10y^{-1}$, $K^{40}/K=1.19\times10^{-4}$ mole/mole

of the *K-Ar* age determinations are given in Table 2. Experimental procedures for *Ar* and *K* analyses are the same as those described elsewhere (*Ozima et al*, 1967).

Because of high air contamination in the determination of radiogenic argon-40 and also of the poor reproducibility in the ages of the Hanamagari lava flows, the values of the *K*-Ar ages should be taken with some reservation. Hence, it is not possible to identify, using only the present results, to which normal event or epoch the Hanamagari lava flows belong. However, it seems to be at least certain that there is a large time span between the Hanamagari and the Kirizumi groups. The latter conclusion is in good accordance with the geological evidences described in the foregoing.

In the case of the Kirizumi lava flows, *K-Ar* ages can be considered to be internally consistent within a flow and between the flows within the experimental error, the latter being estimated to be about 0.15 m.y. at 3 m.y. Hence, we conclude that the *K-Ar* ages of the Kirizumi flows represents the time when the flows were erupted. Considering the paleomagnetic results mentioned above and also the normal-reverse geomagnetic polarity time table proposed by *Dalrymple et al* (1967) and by *McDougall* and *Chamalaun* (1966), it is most likely that the Kirizumi flows were erupted during the Mammoth reversed polarity event.

References

Aramaki, S., Geology of Asama volcano, *Jour. Fac. Sci. Univ. Tokyo*, Sec. II, **14**, 229-443, 1963.

Dalrymple, G.B,. A. Cox, R.R. Doell and C.S. Gromme, Pliocene geomagnetic polarity epochs, *Earth and Planetary Sciences Letters*, **2**, 163-173, 1967.

Iijima, N., Taguchi, K., Ishiwa, K., Koda, M., Nakamura J., Kigune, K., Kobayashi, M., Yano, K. and Yamagishi, I., The volcanoes and the basement of the eastern part of Fossa Magna (in Japanese). *Jour. Assoc. Geol. Collaboration Japan*, No, 62, 622-635, 1958.

McDougall, I. and F.H. Chamalaun, Geomagnetic polarity scale of time, *Nature*, **212**, 1415-1418, 1966.

Ozima, M., M. Kono, I. Kaneoka, H. Kinoshita, K. Kobayashi, T. Nagata, E.E. Larson and D. Strangway, Paleomagnetism and potassium-argon dating of some volcanic rocks from the Rio Grande Gorge, New Mexico, *Jour. Geophys. Res.*, **72**,2615-2621, 1967.

Appendix

Petrographic description of lavas of the Hanamagari and the Kirizumi groups

		KH04	KH06	KH10
phenocryst	plagioclase	⧺slight alteration	⧺partly altered to clay mineral, carbonate etc	⧺slight alteration
	hypersthene	⧺fresh	+fresh	⧺fresh
	augite	⧺fresh	+fresh	⧺fresh
	opaque mineral	+oxide coating	+	+
groundmass	texture	coarse-grained intersertal	fine-grained pilotaxitic	coarse-grained intersertal
	plagioclase	⧺fresh	⧺fresh?	⧺fresh
	orthopyroxene	⧺fresh	+fresh?	?
	clinopyroxene	⧺fresh	+fresh?	⧺fresh
	silica-feldspar mesostasis	⧺	⧺?	⧺
	opaque mineral	+	+	⧺
	clay mineral	+	?	+

⧺ most abundant, ⧺ abundant, + less abundant

Minoru OZIMA, I. KANEOKA, M. KONO, H. KINOSHITA,
K. KOBAYASHI, Y. OHNAKA, T. NAGATA
Geophysical Institute, University of Tokyo
and
S. ARAMAKI
Earthquake Research Institute, University of Tokyo